VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi-590018, Karnataka, INDIA

Industry Internship Report
On

"Data Science with Python"

Submitted By:

Mr. Mohammed Shakir
4AI22CD036

Industry Internship Carried Out at:

Take It Smart (OPC) PVT LTD
Bangalore

Internal Guide

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2025-26

ADICHUNCHANAGIRI INSTITUTE OF TECHNOLOGY

(Affiliated to Visvesvaraya Technological University, Belagavi)
CHIKKAMAGALURU-577 102

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (DATA SCIENCE)

CERTIFICATE

This is to certify that the **Mr. Mohammed Shakir (4AI22CD036)** has successfully completed industry internship **BINT803B (February-2026 to May-2026)** at **Take It Smart (OPC) Pvt. Ltd,** Bangalore. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Internship report has been approved as it satisfies the academic requirements in respect of work prescribed for the award of degree of **Bachelor of Engineering** in **Computer Science & Engineering (Data Science)** of the Visvesvaraya Technological University, Belagavi during the year **2025-2026.**

Signature of the Guide **Mrs. Shilpa K V** B.E., M.Tech **Asst. Professor**	**Signature of the Coordinator** **Ms. Harshitha H D** B.E., M.Tech **Asst. Professor**
Signature of the HOD **Dr. Adarsh M J** B.E., M.Tech., Ph.D **Head of the Dept.**	**Signature of the Principal** **Dr. Goutham M A,** Ph.D **Principal, A I T,** **Chikkamagaluru**

External Examiners

Name of the Examiners | Signature with Date

1. ______________________ ______________________

2. ______________________ ______________________

Consent of Internship Mentor/Supervisor

I having designation Assistant Professor nominated as mentor/supervisor in the organization **Adichunchanagiri Institute of Technology** hereby extend my consent to allow the student Mr. **Mohammed Shakir** of semester 8^{th} USN **4AI22CD036** of Computer Science & Engineering (Data Science) Department to do the internship in **Take It Smart (OPC) Pvt Ltd** organization during the period **Jan 2026-May 2026** Ms. **HARSHITHA H D** or myself will act as an Internship Mentor.

Internship Mentor Signature

Name: Harshitha H D
Designation: Assistant Professor
Address: AIT, Chikkamagaluru
Email: harshithahd@gmail.com

Seal of the Organization

Internship Completion Certificate

It is Certified that Mr. Mohammed Shakir S/O Mohammed Haneef **Department of Computer Science and Engineering (Data Science)** bearing USN 4AI22CD036 of **Adichunchanagiri Institute of Technology,** Chikkamagaluru carried out her internship from 2 Feb 2026 to 16 May 2026 in this organization TAKE IT SMART(OPC) PVT LTD on the bases of her regularity, punctuality, interest shown towards learning skills, team participation, Work experience and meeting Internship objectives, a score of _____marks out of 100 marks is awarded.

Remarks, if any...

Date: 15-05-2026

Signature of Mentor

Name: Mr. Mallikarjun Kumbar

Designation: Director

Address: Take It Smart (OPC) Pvt. Ltd

Email: info@takeitsmart.in

Seal of the Organization

CERTIFICATE OF INTERNSHIP

#startupindia

TAKE IT SMART (OPC) PVT LTD

"Recognized by the Government of India's Startup India Initiative"

MSME NO : UDYAM-KR-03-0236520 | ISO 9001 : 2015 | REG NO : 150027

Internship Certificate

CER ID : TIS-2025-MN15579035

This certifies that

MOHAMMED SHAKIR

STUDENT BEARING USN **4AI22CD036** OF
ADHICHUNCHANAGIRI INSTITUTE OF TECHNOLOGY Has
Successfully Completed Internship In **Data Science** From
2nd February 2026 to 16th May 2026

During This Period, the Intern Exhibited Keen Interest, Professionalism, and willingness to Learn, Fulfilling all assigned Responsibilities to our Satisfaction.
We commend their performance and wish them all the best for their future career.

We extend our best wishes for her/his continued success in both her/his personal and professional endeavors.

TAKE IT SMART (OPC) PVT. LTD. BENGALURU-64

Mallikarjun Kumbar
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ABSTRACT

The internship at Take It Smart Pvt. Ltd. focused on developing practical knowledge in Data Science, Machine Learning, Natural Language Processing (NLP), and Data Analytics. During the internship, I learned Python, SQL, Power BI, Machine Learning concepts, data visualization, and sentiment analysis techniques. The training provided both theoretical understanding and hands-on implementation through coding practice, data preprocessing, analysis, dashboard creation, and project development. It also improved my analytical thinking and problem-solving abilities.

The internship contributed to my professional and personal growth through assignments, presentations, evaluations, and project activities. I worked on projects such as Pizza Sales Analysis and Ecommerce Review Sentiment Analysis, where I applied concepts related to business analytics, dashboard development, NLP, sentiment classification, and predictive analysis. These experiences improved my communication skills, confidence, teamwork, and practical understanding of real-world industry workflows, helping me prepare for future career opportunities in Data Science, Machine Learning, and Artificial Intelligence.

ACKNOWLEDGEMENTS

I express my humble Pranamas to his holiness **Parama Poojya Jagadguru Padmabushana Sri Sri Sri Dr. Balagangadharanatha Mahaswamiji** and **Parama Poojya Jagadguru Sri Sri Sri Dr. Nirmalanandanatha Mahaswamiji** and also to **Sri Sri Gunanatha Swamiji** Sringeri Branch Chikkamagaluru who have showered their blessings on us for framing our career successfully.

I'm deeply indebted to our honorable Director **Dr. C K Subbaraya** for creating the right kind of care and ambience.

I'm thankful to our beloved Principal **Dr. Goutham M A** for inspiring us to achieve greater endeavors in all aspects of Learning.

I express my deepest gratitude to **Dr. Adarsh M J,** Associate Professor & Head, Department of CS&E (DATA SCIENCE) for his valuable guidance, suggestions and constant encouragement without which success of our internship would have been difficult.

I would like to express my sincere gratitude to my internship mentor, **Mrs. Preetha G** for their invaluable guidance, support, and encouragement throughout my internship.

I'm grateful to our department coordinator, **Prof. Harshitha H. D**., for her continuous monitoring and support, and I would also like to thank the college coordinator, **Prof. Manu S.S**, for his guidance and continuous support.

I would like to thank my beloved parents for their support, encouragement and blessings.

And last but not least, I would be pleased to express my heartfelt thanks to all teaching and non-teaching staff of CS&E (DATA SCIENCE) Department and our friends who have rendered their help, motivation and support.

Mohammed Shakir
4AI22CD036

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Data Science is an emerging and highly important field that deals with extracting useful information and meaningful knowledge from large amounts of data. It integrates concepts from statistics, mathematics, computer science, machine learning, and artificial intelligence to analyze data and support effective decision-making in various industries and organizations.

In the modern digital era, enormous volumes of data are continuously produced through platforms such as social media, healthcare systems, educational institutions, banking services, and online businesses. Handling such massive data manually becomes complex and time-consuming; therefore, Data Science provides advanced methods, tools, and technologies to organize, process, and analyze data efficiently while discovering hidden patterns and valuable insights.

The Data Science workflow consists of multiple stages including data collection, data preprocessing, data cleaning, exploratory data analysis, feature engineering, model development, evaluation, and deployment. Data can be gathered from several sources such as databases, websites, APIs, spreadsheets, and cloud-based systems. Data preprocessing and cleaning are essential steps that improve data quality by removing duplicate entries, handling missing values, and correcting inconsistencies. Exploratory Data Analysis (EDA) is performed to better understand the structure and characteristics of data using statistical techniques and visual representations.

Data visualization is another significant aspect of Data Science, as it helps represent information in a clear and understandable format through graphs, charts, dashboards, and reports. Visualization tools and libraries such as Matplotlib, Seaborn, Tableau, and Power BI are widely used for presenting analytical results effectively. Furthermore, Machine Learning algorithms are applied for tasks such as prediction, classification, automation, and recommendation systems in real-world applications.

Various technologies and programming tools are utilized in the field of Data Science, including Python, R programming, SQL, Excel, Hadoop, and Apache Spark.

Among these, Python is one of the most commonly used programming languages because of its easy syntax and extensive libraries such as NumPy, Pandas, Scikit-learn, TensorFlow, and Keras. SQL is primarily used for managing databases and performing data querying operations efficiently.

Data Science has applications across several domains including healthcare, finance, e-commerce, transportation, cybersecurity, and education. Organizations use Data Science techniques to make informed decisions, improve operational efficiency, minimize costs, and enhance customer experiences. It also plays a crucial role in supporting modern technologies such as Artificial Intelligence (AI), Deep Learning, Cloud Computing, Big Data Analytics, and Business Intelligence systems. These technologies help businesses automate operations, predict future trends, understand customer preferences, and optimize organizational performance. Recommendation systems used in online shopping platforms and entertainment services are among the popular applications of Data Science.

Practical implementation is an essential part of learning Data Science concepts effectively. Working on real-time projects helps in developing analytical thinking, technical expertise, and problem-solving skills. Handling datasets and building machine learning models provide practical understanding of predictive analytics, visualization methods, and business intelligence solutions.

As a part of academic learning and industrial training, an internship was completed at Take It Smart Pvt. Ltd. The internship offered hands-on experience in areas such as Python programming, SQL, Machine Learning, Data Analysis, Power BI, and Streamlit application development, along with exposure to real-time industry projects and professional practices.

During the internship period, several projects including Pizza Sales Analysis, Ecommerce Review Sentiment Analysis, Student Performance Prediction, and House Price Prediction were developed using Data Science and Machine Learning methodologies. These projects provided practical exposure to data preprocessing, dashboard development, data visualization, predictive modeling, and analytical techniques through real-world implementation.

In summary, Data Science has become one of the most valuable technologies for transforming raw data into meaningful information and intelligent solutions. With the continuous advancement of Artificial Intelligence, automation, cloud technologies, and Big Data, the demand and significance of Data Science continue to grow across industries, creating new opportunities for innovation, research, and business development worldwide.

1.2 About the Organization

Take It Smart Pvt. Ltd. is a technology-oriented organization that delivers services and training in domains such as software development, Data Science, Artificial Intelligence, and other emerging digital technologies. The organization aims to support students and working professionals in developing practical skills through internship programs, technical training sessions, and exposure to real-time industrial projects.

The company provides learners with opportunities to participate in project-based learning, which helps in enhancing both technical knowledge and professional competencies. Through proper mentorship, guidance, and practical implementation, students are able to understand modern technologies and their real-world applications in the fields of Data Science and Machine Learning.

The organization also maintains a professional and supportive learning atmosphere where interns can strengthen their technical capabilities, gain hands-on industrial experience, and become familiar with professional work culture, collaboration, and teamwork practices followed in the industry.

1.2.1 Key Highlights of the Organization

- Focuses on advanced technologies including Data Science, Machine Learning, Artificial Intelligence, and Software Development.
- Conducts internship programs and professional technical training sessions for students and beginners to enhance their industry knowledge.
- Emphasizes practical learning by involving trainees in real-time projects and hands-on implementation activities.
- Provides proper mentorship, technical support, and guidance from skilled industry experts and professionals.
- Helps learners improve their skills in Python programming, SQL, Data Analytics, and other modern technologies related to Data Science.

CHAPTER 2

Objectives of Internship

2.1 Objectives

- To gain practical knowledge and hands-on experience in Data Science and Machine Learning.
- To understand the Data Science lifecycle including data collection, preprocessing, analysis, visualization, and model development.
- To improve Python programming skills using libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.
- To perform Exploratory Data Analysis (EDA) for identifying insights, trends, and patterns in datasets.
- To build and evaluate Machine Learning models using different algorithms and evaluation techniques.
- To enhance technical, analytical, SQL, and problem-solving skills through practical project implementation.

2.2 Technical Objectives

The technical objectives of the internship were comprehensive and covered major areas of Data Science, Machine Learning, SQL, Data Analytics, and Business Intelligence. Each objective was designed to improve practical knowledge and technical skills through progressive learning, real-world project implementation, data analysis, visualization, and machine learning model development.

- **Python for Data Science**

Python is one of the most widely used programming languages in the field of Data Science and Machine Learning because of its simplicity, flexibility, and extensive collection of powerful libraries. It is commonly applied in areas such as data analysis, visualization, machine learning, deep learning, and automation. During the internship period, Python was mainly utilized for data preprocessing, Exploratory Data Analysis (EDA), machine learning model creation, and deep learning applications.

The Python libraries used during the internship are:

- **NumPy:** NumPy is a numerical computing library used for mathematical calculations and efficient handling of multidimensional arrays. It supports operations such as array manipulation, indexing, reshaping, and numerical computations.
- **Pandas:** Pandas is a data manipulation and analysis library used for handling structured datasets. It helps in tasks such as data cleaning, handling missing values, filtering, grouping, and analyzing data efficiently.
- **Matplotlib / Seaborn:** Matplotlib and Seaborn are data visualization libraries used to create graphs and charts such as histograms, scatter plots, line charts, box plots, heatmaps, and count plots to understand patterns and relationships in data.
- **Scikit-learn:** Scikit-learn is a machine learning library used for building and evaluating models. It provides algorithms for regression, classification, clustering, preprocessing, feature scaling, and model evaluation.
- **TensorFlow / Keras:** TensorFlow and Keras are deep learning frameworks used to build Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN) for applications such as image classification and prediction systems.
- **Plotly:** Plotly is an interactive visualization library used for creating dynamic graphs, dashboards, and visual reports for better data analysis and presentation,

SQL for Data Science:

SQL (Structured Query Language) is a database language used for storing, retrieving, and managing data in relational database systems. In Data Science, SQL plays an important role in extracting and analyzing large datasets before applying machine learning techniques. Since organizational information is generally stored in tabular format, SQL helps in efficiently organizing, filtering, sorting, and analyzing data.

Common SQL operations used include:

- **SELECT** – used to retrieve data from database tables
- **WHERE** – used to filter records based on specific conditions

- **JOIN** – used to combine data from multiple related t
- **GROUP BY** – used to group and summarize similar records

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is an essential step in Data Science used to understand datasets before developing machine learning models. It helps in identifying missing values, detecting outliers, understanding data distributions, and analyzing relationships between features.

Objectives of Exploratory Data Analysis:

- To understand dataset structure and feature types
- To identify missing values and duplicate records
- To detect inconsistencies and outliers
- To analyze trends, patterns, and feature relationships
- To prepare cleaned data for machine learning models

Steps in Exploratory Data Analysis:

- **Data Collection:** Datasets were collected from CSV files, Kaggle, and real-world project sources for analysis and model development.
- **Data Cleaning:** Missing values were handled, duplicate records were removed, and inconsistencies were corrected to improve data quality.
- **Data Understanding:** Functions such as head(), info(), describe(), shape(), and value_counts() were used to study dataset structure and statistical details.
- **Data Visualization:** Libraries such as Matplotlib, Seaborn, and Plotly were used to create graphs and charts for identifying trends and insights.

Power BI for Data Science

Power BI is a Business Intelligence tool used to create interactive dashboards and visual reports. It helps convert raw data into meaningful insights using charts, graphs, and visual analytics.

In Data Science, Power BI is mainly used after data analysis and model development to present results clearly and support better decision-making.

Objectives:

- To represent data visually in an understandable format
- To create interactive dashboards and analytical reports
- To identify trends, insights, and patterns within data
- To support decision-making using data-driven approaches

Machine Learning

Machine Learning is a branch of Artificial Intelligence that enables systems to learn from data and make predictions without explicit programming. It identifies patterns within datasets and improves model performance through training.

During the internship, algorithms such as Linear Regression, Logistic Regression, Decision Tree, Random Forest, KNN, SVM, XGBoost, and K-Means Clustering were implemented for tasks related to prediction, classification, and clustering using various datasets.

Deep Learning

Deep Learning is a subset of Machine Learning that uses neural networks with multiple hidden layers to learn complex patterns from large datasets. During the internship, deep learning techniques were implemented for applications such as Face Mask Detection and Digit Recognition.

Concepts including Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), activation functions such as ReLU, Sigmoid, and Softmax, forward and backward propagation, optimizers like Adam and Gradient Descent, image preprocessing, normalization, and data augmentation were studied and implemented.

2.3 Practical and Professional Objectives

The internship emphasized that technical knowledge, practical experience, and professional skills are essential for a successful career in Data Science and the IT industry. Therefore, it focused on improving practical, analytical, communication, and teamwork skills through real-world projects and hands-on learning.

Practical Objectives

- To obtain hands-on experience in Data Science, Machine Learning, and Deep Learning using real-world datasets and projects.
- To apply theoretical concepts to practical industry-oriented tasks and project implementation.
- To understand the complete workflow starting from data collection and preprocessing to model development, evaluation, and deployment.
- To enhance programming skills using Python, SQL, and Data Science-related libraries and frameworks.
- To perform Exploratory Data Analysis (EDA) and generate valuable insights from datasets.
- To develop regression, classification, clustering, and deep learning models for solving practical business and analytical problems.
- To gain practical experience in deploying Machine Learning applications using Streamlit technology.

Professional Objectives

- To improve professional communication and technical documentation skills.
- To develop time management and task-handling abilities during project execution.
- To enhance presentation and reporting skills through internship documentation and reports.
- To gain confidence in handling real-world Data Science and Machine Learning projects independently.
- To improve teamwork and collaborative learning abilities during internship activities.
- To prepare for future career opportunities in Data Science, Artificial .

CHAPTER 3

LEARNING EXPERIENCES

3.1 Introduction to Learning Experience

The internship carried out at Take It Smart Pvt. Ltd. provided valuable practical exposure to different concepts related to Data Science, Machine Learning, Deep Learning, SQL, and Artificial Intelligence. The internship created an opportunity to understand how theoretical concepts learned academically can be applied in real-time industrial projects and business-oriented applications.

During the internship period, I worked on multiple stages of the Data Science lifecycle, including data collection, preprocessing, data cleaning, exploratory data analysis, visualization, model development, evaluation, and deployment. Working with different datasets helped me understand the process of transforming raw data into meaningful insights and applying analytical techniques for prediction and decision-making tasks.

I improved my programming skills in Python and gained practical experience using libraries such as NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, and Plotly. In addition, I developed knowledge in SQL and PostgreSQL for database management, data querying, and structured data analysis.

During the internship, I worked on projects such as Pizza Sales Analysis and Ecommerce Review Sentiment Analysis. These projects provided practical understanding of business analytics, dashboard creation, Natural Language Processing (NLP), sentiment classification, and predictive analytics using real-world datasets and modern technologies.

Overall, the internship helped me gain practical knowledge, improve analytical and problem-solving abilities, and build confidence in handling real-world Data Science andMachine Learning applications. It also provided industry exposure and improved my fields in data science , machine learning.

3.2 Technical Learning

Through the internship at Take It Smart Pvt. Ltd., I gained hands-on experience and technical knowledge in areas such as Data Science, Machine Learning, Deep Learning, SQL, Data Analytics, and Natural Language Processing. The internship helped me understand the complete workflow involved in developing Data Science solutions, beginning from data collection and preprocessing to model building and deployment.

I strengthened my programming knowledge in Python and worked with libraries such as NumPy and Pandas for tasks including data preprocessing, cleaning, manipulation, and analysis. I learned methods for handling missing values, removing duplicate records, encoding categorical variables, normalization, and feature scaling techniques.

The internship also improved my understanding of Exploratory Data Analysis (EDA) using visualization libraries such as Matplotlib, Seaborn, and Plotly. I analyzed datasets using visualizations including histograms, scatter plots, line charts, heatmaps, and box plots to identify trends, relationships, and patterns within the data.

I gained practical exposure in implementing Machine Learning algorithms using Scikit-learn. I worked with supervised learning algorithms such as Linear Regression, Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Gradient Boosting, and XGBoost for regression and classification tasks. I also explored unsupervised learning techniques including K-Means Clustering for grouping and pattern analysis.

In addition, I studied Deep Learning concepts using TensorFlow and Keras frameworks. Topics such as Artificial Neural Networks (ANN), backward propagation, normalization, and data augmentation techniques were learnedand implemented during the internship.

The internship also provided practical experience in SQL and PostgreSQL using pgAdmin, where I practiced creating databases and tables, writing SQL queries, performing joins, aggregations, group subqueries, CASE statements, ranking functions, window functions, and Common Table Expressions (CTE).

I learned various model evaluation methods using metrics such as Accuracy, Precision, Recall, F1-Score, MAE, MSE, RMSE, R^2 Score, and Confusion Matrix for evaluating machine learning model performance. Additionally, I gained practical knowledge in deploying machine learning applications using Streamlit for developing real-time prediction and sentiment analysis systems.

As part of the internship projects, I implemented Pizza Sales Analysis using Python, SQL (SSMS), and Power BI for business analytics and dashboard visualization. I also developed an Ecommerce Review Sentiment Analysis project using Python, Transformers and Streamlit for customer review classification and sentiment prediction. These projects enhanced my understanding of predictive analytics, visualization techniques, NLP concepts, and real-world business problem solving.

Overall, the internship significantly strengthened my technical, analytical, and problem-solving abilities in Data Science, Machine Learning, Deep Learning, SQL, Data Visualization, and Predictive Analytics. It also improved my confidence in handling real-world datasets and implementing practical solutions using modern tools and technologies.

3.3 Practical Implementation

During the internship at Take It Smart Pvt. Ltd., I gained hands-on experience by working on real-world datasets and implementing concepts related to Data Science, Machine Learning, Deep Learning, and SQL. The internship provided practical exposure to the complete Data Science workflow, including data collection, preprocessing, analysis, model development, evaluation, deployment, and result interpretation.

I worked extensively with Python and libraries such as NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Plotly, and Transformers for performing data analysis, machine learning, deep learning, and visualization.

One of the major projects implemented during the internship was the Pizza

Analysis project. In this project, pizza sales datasets were analyzed using Python, SQL Server Management Studio (SSMS), and Power BI. SQL queries were used to perform KPI calculations and analyze sales-related information such as total revenue, total orders, average order value, sales trends, top-selling pizzas, and customer preferences. Exploratory Data Analysis (EDA) was performed using Python libraries, and interactive dashboards were created in Power BI to visualize business insights effectively. This project improved my understanding of business analytics, dashboard development, data visualization, and real-world decision-making systems.

Another important project completed during the internship was the Ecommerce Review Sentiment Analysis project. In this project, customer reviews from ecommerce platforms were analyzed using Natural Language Processing (NLP) techniques. A pre-trained Transformer-based model, DistilBERT, from the Transformers library was used for sentiment classification. The system was designed to classify reviews as positive or negative and generate confidence scores for predictions. Streamlit was used to develop an interactive web application for real-time sentiment analysis and visualization of sentiment distribution. This project helped me understand NLP concepts, text classification, sentiment analysis, and deployment of machine learning applications.

During the internship, I performed Exploratory Data Analysis (EDA) on multiple datasets to identify patterns, trends, correlations, and outliers. Visualization techniques such as histograms, scatter plots, heatmaps, bar charts, box plots, and line graphs were used to generate meaningful insights and improve understanding of data behavior.

I also implemented several Machine Learning algorithms including Linear Regression, Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN),

The internship additionally provided practical exposure to Deep Learning concepts using Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN). I learned image preprocessing techniques such as resizing, normalization, and data augmentation for image classification applications.

In addition, I gained practical experience in SQL and PostgreSQL using pgAdmin for database management and querying operations. I practiced database creation, filtering, joins, aggregation functions, grouping, subqueries, CASE statements, ranking functions, window functions, and Common Table Expressions (CTE) for structured data analysis.

I also learned various model evaluation techniques using metrics such as Accuracy, Precision, Recall, F1-Score, MAE, MSE, RMSE, R^2 Score, and Confusion Matrix for measuring machine learning model performance. Furthermore, I gained experience in deploying Machine Learning applications using Streamlit for developing interactive prediction and analytics systems.

Overall, the practical implementation experience significantly improved my programming, analytical thinking, debugging, and problem-solving abilities. It also provided a better understanding of how Data Science, Machine Learning, NLP, SQL, and Business Intelligence techniques are applied to solve real-world business and analytical problems across different domains.

3.4 Project Experience

During the internship, I worked on multiple Data Science and Machine Learning projects that improved my practical understanding and technical knowledge. These projects provided hands-on experience in data preprocessing, exploratory data analysis, machine learning model implementation, visualization, dashboard creation, Natural Language Processing (NLP), and deployment using real-world datasets and industry-oriented methodologies.

- **Ecommerce Review Sentiment Analysis using Python**

The Ecommerce Review Sentiment Analysis project was developed to analyze customer feedback and classify reviews based on sentiment using Natural Language Processing and Machine Learning techniques. The primary objective of this project was to build an intelligent sentiment analysis system capable of identifying whether customer reviews were positive or negative and generating meaningful insights from textual data.

The project started with loading and understanding the ecommerce review dataset using Pandas. I performed preprocessing tasks such as handling missing values, cleaning textual data, removing duplicate records, and preparing review text for sentiment analysis. Exploratory Data Analysis (EDA) was also conducted to understand review distributions and sentiment patterns using visualization techniques and statistical analysis.

For sentiment classification, I implemented a Transformer-based Natural Language Processing model using DistilBERT from the Transformers library

The system was tested by providing customer reviews as input and generating sentiment labels such as positive or negative as output. Streamlit was used to build an interactive web application where users could analyze individual reviews in real time and visualize overall sentiment distribution using charts and graphs.

The project improved my understanding of Natural Language Processing, sentiment analysis, text preprocessing, Transformer models, and deployment of machine learning applications. It also helped me understand how sentiment analysis systems are used in ecommerce platforms and business environments to analyze customer opinions, improve products and services, and support business decision-making.

- **Pizza Sales Analysis using Python, SQL, and Power BI**

The Pizza Sales Analysis project was developed to analyze sales performance and customer ordering patterns using Python, SQL, and Power BI. The main objective of this project was to extract meaningful business insights from pizza sales data and support better decision-making through data analysis and visualization techniques.

In the project, I worked with sales datasets containing information such as order dates, pizza categories, pizza sizes, quantities sold, prices, and total revenue. I performed data preprocessing tasks including handling missing values, removing inconsistencies, and formatting date-related information for analysis.

I conducted Exploratory Data Analysis (EDA) to identify sales trends, peak ordering periods, customer preferences, and high-performing pizza categories.

Various visualization techniques such as bar charts, line graphs, pie charts, histograms, and KPI cards were used to understand revenue distribution, order patterns, and sales performance.

Using SQL, I wrote queries to calculate important business metrics such as total revenue, total orders, total pizzas sold, average order value, and average pizzas per order. I also analyzed daily and monthly sales trends and identified top-performing and low-performing pizzas based on revenue, quantity sold, and number of orders.

The dataset was further connected to Power BI for dashboard development. I created interactive dashboards using charts, slicers, filters, and KPI indicators to provide clear insights into business performance. The dashboard helped in analyzing customer behavior, sales growth, and product performance effectively.

The project improved my understanding of data preprocessing, business analytics, SQL querying, dashboard development, and data visualization techniques. It also helped me understand how data analysis can support business decision-making, improve sales strategies, and identify customer preferences in the food and retail industry.

Overall, the project strengthened my practical skills in Python, SQL, Power BI, and data analytics while increasing my confidence in handling real-world business datasets and analytical tasks.

3.5 Teamwork & Professional Learning

During the internship at Take It Smart Pvt. Ltd., I gained both technical and professional learning experience in the areas of Data Science, Machine Learning, SQL, Business Intelligence, and Natural Language Processing. The internship helped me understand the importance of communication, teamwork, discipline, adaptability, and time management in a professional working environment.

Throughout the internship, I actively participated in project discussions, assignments, technical sessions, and practical implementations related to Python, Machine Learning, SQL, Power BI, Streamlit, and dashboard development. Regular interaction with mentors and trainers improved my communication .

Working on projects such as Pizza Sales Analysis and Ecommerce Review Sentiment Analysis improved my ability to analyze problems and develop practical solutions using real-world datasets and modern technologies. I also learned the importance of collaboration, knowledge sharing, debugging, and teamwork while discussing project implementation, improving visualizations, analyzing datasets, and enhancing model performance.

The internship strengthened my professional skills by teaching me how to manage tasks within deadlines, prepare technical documentation, present project results clearly, and organize project workflows efficiently. I also gained exposure to industry practices, reporting methods, and development processes followed in Data Science, analytics, and Machine Learning-based projects.

In addition, the internship improved my confidence in handling real-world datasets independently and strengthened my analytical thinking, adaptability, and problem-solving abilities. Overall, the experience enhanced both my technical expertise and professional readiness for future opportunities in Data Science, Machine Learning, Artificial Intelligence, and Data Analytics.

3.6 Challenges

During the internship at Take It Smart Pvt. Ltd., I encountered several challenges while learning and implementing concepts related to Data Science, Machine Learning, SQL, Power BI, Natural Language Processing, and Deep Learning. These challenges helped me improve my technical understanding, logical thinking, and practical problem-solving abilities.

One of the major challenges was understanding advanced Machine Learning concepts such as regression, classification, clustering, feature engineering, model evaluation, and optimization techniques. Initially, implementing algorithms and interpreting prediction results was difficult, but regular practice and project implementation improved my understanding gradually.

Handling real-world datasets was another challenge because many datasets contained missing values, duplicate records, inconsistent formatting, and outliers. I learned different preprocessing techniques to clean, transform, and prepare datasets for analysis and machine learning model development.

While working on the Pizza Sales Analysis project, I faced challenges in KPI calculation, SQL querying, dashboard creation, and identifying meaningful business insights from large sales datasets. Understanding sales trends, customer ordering behavior, and selecting suitable visualizations in Power BI required detailed analysis and experimentation.

Similarly, during the Ecommerce Review Sentiment Analysis project, I faced difficulties in text preprocessing, understanding Natural Language Processing concepts, implementing Transformer-based models, and interpreting sentiment prediction results. Learning how DistilBERT processes text data and generates sentiment classification outputs required additional experimentation and study.

Learning advanced SQL concepts such as joins, subqueries, window functions, Common Table Expressions (CTE), aggregation functions, and ranking operations was challenging initially. Building interactive dashboards in Power BI and selecting appropriate charts and KPI indicators for business analysis also required continuous practice and improvement.

I also faced challenges while learning Deep Learning concepts including Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), TensorFlow, and Keras implementation. Understanding neural network training, parameter tuning, activation functions, and improving model performance required additional guidance and experimentation.

Managing multiple assignments, practical tasks, projects, and report preparation within deadlines was another challenge during the internship. However, this experience helped me improve my time management, organizational, and multitasking abilities.

Despite these challenges, the internship provided valuable practical exposure and learning opportunities that strengthened my confidence and helped me develop industry-oriented skills in Data Science, Machine Learning, SQL, Business Analytics, NLP, and Artificial Intelligence. It also improved my ability to handle real-world datasets, solve technical problems, and work effectively on practical projects and analytical tasks.

3.7 Project Overview

During the internship, I worked on multiple Data Science, Machine Learning, and Business Analytics projects that provided practical exposure to real-world data analysis, visualization, Natural Language Processing, and predictive analytics applications. These projects involved different stages of the Data Science workflow including data preprocessing, Exploratory Data Analysis (EDA), visualization, model development, evaluation, dashboard creation, and deployment using real-world datasets.

One of the major projects completed during the internship was Pizza Sales Analysis using Python, SQL, and Power BI, where sales data was analyzed to identify customer ordering patterns, revenue trends, best-selling pizzas, and overall business performance. The project involved data cleaning, KPI calculation, SQL querying, trend analysis, and interactive dashboard development using Power BI.

Another important project was Ecommerce Review Sentiment Analysis using Python, Transformers, and Streamlit, which focused on analyzing customer reviews and classifying sentiments as positive or negative using Natural Language Processing techniques. The project involved text preprocessing, sentiment classification using DistilBERT, real-time prediction, and interactive visualization using Streamlit.

These projects improved my understanding of machine learning concepts, business analytics, Natural Language Processing, dashboard development, data visualization, and predictive analytics through practical implementation.

Overall, the projects strengthened my technical skills in Python, SQL, Power BI, Streamlit, Data Visualization, Machine Learning, NLP, and Business Analytics while improving my analytical thinking, problem-solving abilities, and confidence in handling real-world datasets and applications.

3.7.1 Introduction

During the internship at Take It Smart Pvt. Ltd., I worked on multiple Data Science and Machine Learning projects that provided practical exposure to solving real-world business and analytical problems using modern technologies and predictive analytics techniques. The major projects completed during the internship were Pizza Sales Analysis using Python, SQL, and Power BI and Ecommerce Review Sentiment.

3.7.2 Introduction

During the internship at Take It Smart Pvt. Ltd., I worked on multiple Data Science and Machine Learning projects that provided practical exposure to solving real-world business and analytical problems using modern technologies and predictive analytics techniques. The major projects completed during the internship were Pizza Sales Analysis using Python, SQL, and Power BI and Ecommerce Review Sentiment Analysis using Python, Transformers, and Streamlit. These projects involved complete workflows including data preprocessing, Exploratory Data Analysis (EDA), visualization, machine learning implementation, evaluation, and deployment.

3.7.3 Problem Statement

Real-world datasets are often large, unstructured, and contain inconsistencies that make manual analysis difficult and time-consuming. The objective of these projects was to apply Data Science, SQL, Machine Learning, NLP, and visualization techniques to analyze data, identify meaningful patterns, and generate useful business insights and sentiment predictions.

Pizza Sales Analysis:

Analyze pizza sales data to identify customer ordering patterns, revenue trends, best-selling pizzas, and business performance using Python, SQL, and Power BI.

Ecommerce Review Sentiment Analysis:

Develop a sentiment analysis system capable of classifying ecommerce customer reviews as positive or negative using Natural Language Processing and Transformer-based Machine Learning techniques.

Other Projects:

Student performance analysis, HR analytics, and dashboard-based business analysis using visualization tools and machine learning techniques.

3.7.4 Objectives

- To understand the complete workflow of Data Science and Machine Learning projects.
- To perform data preprocessing and Exploratory Data Analysis (EDA) on
- structured and textual datasets.

- To develop business analytics dashboards and recommendation systems using real-world datasets.
- To implement Machine Learning and NLP techniques for prediction and sentim -ent classification tasks.
- To analyze datasets using SQL queries and visualization techniques.
- To deploy Machine Learning applications using Streamlit for interactive user experience.

3.7.5 Dataset Description

Pizza Sales Dataset:

The dataset included information such as pizza category, pizza size, quantity sold, order date, total price, and customer order details used for sales analysis and KPI generation.

Ecommerce Review Dataset:

The dataset included customer review text and sentiment-related information used for sentiment classification and analysis using NLP techniques.

Other Datasets:

Student academic details, HR employee data, and business datasets were also used for prediction, analysis, and dashboard creation.

The datasets were preprocessed using techniques such as handling missing values, removing duplicates, encoding categorical variables, and feature transformation before analysis and model development.

3.7.6 System Architecture

The project workflow followed a structured Data Science architecture:

- **Data Collection:** Datasets were collected from CSV files, online sources, and project datasets.
- **Data Preprocessing:** Data cleaning techniques such as handling missing values, removing duplicates, and formatting data were performed.
- **Exploratory Data Analysis (EDA):** Statistical analysis and visualizations were used to identify trends, relationships, and patterns within datasets.
- **Feature Engineering:** Important features were selected and transformed to improve recommendation quality and analysis performance.
- **Model Development:** Machine Learning techniques and analytical methods were implemented for prediction and recommendation tasks.

- **Model Evaluation:** The system performance was analyzed using suitable evaluation methods and business insights.
- **Visualization and Dashboard Development:** Interactive dashboards and charts were created using Power BI and Plotly for better analysis and reporting.
- **Deployment using Streamlit:** The final applications were deployed using Streamlit for interactive usage and real-time output generation.

3.7.7 Tools and Technologies Used

The following tools and technologies were used during project implementation:

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Plotly
- Scikit-learn
- SQL
- PostgreSQL (pgAdmin)
- Power BI
- Streamlit
- Google Colab
- TensorFlow
- Keras
- Transformers (DistilBERT)

3.7.8 Algorithms and Techniques Used

Machine Learning and NLP Techniques

- TF-IDF Vectorization
- Sentiment Classification
- Transformer-based NLP Model (DistilBERT)
- Content Analysis Techniques

Data Analysis Techniques

- Exploratory Data Analysis (EDA)
- KPI Analysis
- Trend Analysis

- Data Visualization

SQL Operations

- SELECT
- JOIN
- GROUP BY
- Subqueries
- Aggregate Functions

3.7.9 Working of the System

The system works by collecting datasets and processing them through preprocessing, cleaning, formatting, and analysis stages. After preprocessing, analytical methods and machine learning techniques are applied to generate insights, recommendations, and visual outputs.

- In the Pizza Sales Analysis project, sales-related data was analyzed to calculate KPIs such as total revenue, total orders, and top-selling pizzas. Interactive dashboards were created to visualize business performance and sales trends.

- In the Ecommerce Review Sentiment Analysis project, customer review text was processed using the DistilBERT Transformer model to classify reviews as positive or negative and generate confidence scores for predictions. Streamlit was used to create an interactive interface for real-time sentiment analysis and visualization.

- Other projects generated outputs such as student performance analysis, HR analytics insights, and dashboard-based business reports.

The final systems generated meaningful insights and predictions based on patterns identified from historical datasets.

3.7.10 Output and Results

The implemented projects successfully generated analytical insights, recommendations, and visualization outputs.

Pizza Sales Analysis

- Analyzed sales trends and customer ordering behavior.
- Calculated KPIs such as total revenue, orders, and average order value.

- Identified best-selling and low-performing pizzas.
- Developed interactive Power BI dashboards for business analysis.

Ecommerce Review Sentiment Analysis

- Classified customer reviews as positive or negative.
- Generated sentiment predictions with confidence scores.
- Developed an interactive Streamlit application for real-time analysis.
- Visualized overall sentiment distribution using charts and graphs.

Other Projects

- Analyzed student performance and HR datasets.
- Created dashboards and visual reports for data interpretation.
- Generated meaningful insights from structured datasets.

3.7.11 Advantages

- Helps in analyzing real-world datasets efficiently.
- Supports data-driven business decision-making and analytics.
- Provides practical exposure to Machine Learning, NLP, and Data Analytics concepts.
- Generates interactive dashboards and meaningful visualizations.
- Improves understanding of customer opinions through sentiment analysis.
- Enhances analytical thinking and problem-solving abilities.

3.7.11 Conclusion

The projects completed during the internship provided valuable practical exposure to Data Science, Machine Learning, SQL, NLP, and Business Analytics concepts. Through the implementation of Pizza Sales Analysis and Ecommerce Review Sentiment Analysis, I gained hands-on experience in preprocessing, Exploratory Data Analysis (EDA), visualization, dashboard development, sentiment classification, and deployment.

The internship improved my programming skills, analytical thinking, and practical understanding of predictive analytics, Natural Language Processing, and business intelligence systems. I also learned how Data Science, Machine Learning, and NLP techniques can be applied to solve real-world business and analytical problems and generate intelligent, data-driven solutions.

CHAPTER 4

LEARNING OUTCOMES

4.1 Technical Learning Outcomes

During the internship at Take It Smart Pvt. Ltd., I gained valuable practical knowledge and technical exposure in the fields of Data Science, Machine Learning, Natural Language Processing, SQL, Data Analytics, and Business Intelligence. The internship enhanced both my theoretical understanding and practical implementation skills through assignments, real-world datasets, and project-based learning.

I improved my programming skills in Python and worked with libraries such as NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, Plotly, and Transformers for tasks related to data preprocessing, analysis, visualization, machine learning, NLP, and dashboard development.

I gained hands-on experience in data preprocessing techniques including handling missing values, removing duplicate records, encoding categorical variables, feature scaling, normalization, text preprocessing, and feature engineering. I also strengthened my understanding of Exploratory Data Analysis (EDA) using visualizations such as histograms, scatter plots, heatmaps, line charts, bar graphs, and box plots to identify trends, correlations, and patterns within datasets.

The internship provided practical exposure to implementing Machine Learning algorithms such as Linear Regression, Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Gradient Boosting, and XGBoost for regression and classification tasks. In addition, I gained knowledge in Natural Language Processing techniques and Transformer-based models through the Ecommerce Review Sentiment Analysis project using DistilBERT.

I also learned Deep Learning concepts including Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), activation functions, normalization, image preprocessing, and model training using TensorFlow and Keras frameworks. Furthermore, I improved my understanding of SQL and PostgreSQL for querying, joins, aggregation functions, subqueries, window functions, Common Table

Expressions (CTE), and database management operations.

The internship also enhanced my skills in creating Power BI dashboards and visual reports for business analytics. I learned model evaluation techniques using metrics such as Accuracy, Precision, Recall, F1-Score, MAE, MSE, RMSE, R^2 Score, and Confusion Matrix for measuring machine learning model performance. Additionally, I gained practical experience in deploying Machine Learning and NLP applications using Streamlit for real-time analysis and prediction systems.

Through projects such as Pizza Sales Analysis and Ecommerce Review Sentiment Analysis, I gained practical understanding of business analytics, dashboard development, sentiment classification, and real-world data analysis techniques.

Overall, the internship significantly strengthened my technical skills in Data Science, Machine Learning, SQL, Power BI, Natural Language Processing, Data Visualization, and Predictive Analytics while improving my confidence in handling real-world datasets and analytical projects.

4.2 Practical Learning Outcomes

During the internship at Take It Smart Pvt. Ltd., I gained practical and professional experience through project implementation, continuous learning, and mentor guidance.

The internship improved my communication and presentation skills through technical discussions, project demonstrations, and documentation work. I learned how to prepare technical reports and explain project concepts clearly.

I also improved my task management and organizational skills by completing assignments, presentations, and project activities within deadlines.

Working with real-world datasets increased my confidence in applying Data Science and Machine Learning techniques to business and analytical problems. I also gained exposure to project workflows including preprocessing, model development, evaluation, reporting, and deployment.

Through the Pizza Sales Analysis project, I learned business analytics and dashboard visualization techniques. The Ecommerce Review Sentiment Analysis project helped me understand real-world NLP and sentiment analysis applications.

The internship improved my analytical thinking, debugging, and problem-solving abilities while working with different models and techniques. It also encouraged continuous self-learning and helped me explore modern tools and technologies beyond academics.

Overall, the internship enhanced my professional behavior, technical confidence, communication, teamwork, and problem-solving skills for future opportunities in Data Science and Machine Learning.

4.3 Professional & Personal Development

During the internship at Take It Smart Pvt. Ltd., I achieved professional and personal growth through project-based learning, teamwork, and mentor guidance in Data Science and Machine Learning.

The internship improved my communication, presentation, and reporting skills through project discussions, technical explanations, and documentation work. I learned how to present analytical results and prepare structured reports.

I also improved my time management and organizational skills by handling assignments, practical tasks, and project work within deadlines. This helped me become more disciplined and responsible in managing professional activities.

Working on projects such as Pizza Sales Analysis and Ecommerce Review Sentiment Analysis improved my confidence in handling real-world datasets and implementing practical Data Science solutions. I also gained knowledge of project workflows including preprocessing, analysis, model development, evaluation, dashboard creation, and deployment.

On a personal level, the internship strengthened my confidence, analytical thinking, adaptability, and independent learning abilities. Solving coding issues and improving model performance helped me enhance my logical thinking and problem-solving skills.

Overall, the internship contributed significantly to my technical confidence, communication skills, teamwork abilities, and professional development while preparing me for future opportunities in Data Science, Machine Learning, and Artificial Intelligence.

CHAPTER 5

ATTACHMENTS

5.1 Internship Work Summary

During my internship at Take It Smart Pvt. Ltd., I gained practical experience in Data Science, Machine Learning, SQL, Data Analytics, NLP, and Business Intelligence through theoretical learning and real-world project implementation.

Initially, I learned Python programming concepts such as variables, loops, functions, lists, tuples, and dictionaries. I also practiced coding and problem-solving exercises to improve my programming and logical thinking skills.

During the internship, I worked with libraries such as NumPy, Pandas, Matplotlib, Seaborn, Plotly, Scikit-learn, TensorFlow, Keras, and Transformers for data preprocessing, visualization, machine learning, and NLP tasks. I learned handling missing values, cleaning datasets, performing Exploratory Data Analysis (EDA), and generating insights from data.

I implemented Machine Learning algorithms including Linear Regression, Logistic Regression, Decision Tree, Random Forest, KNN, SVM, Gradient Boosting, and XGBoost for regression and classification tasks. I also learned evaluation metrics such as Accuracy, Precision, Recall, F1-Score, MAE, MSE, RMSE, and R^2 Score.

During the internship, I completed projects such as Pizza Sales Analysis and Ecommerce Review Sentiment Analysis. These projects helped me understand business analytics, sentiment analysis, dashboard development, and predictive modeling using real-world datasets.

In the Pizza Sales Analysis project, I used Python, SQL (SSMS), and Power BI to analyze sales trends, customer ordering behavior, revenue, and KPIs. Interactive dashboards were created for business analysis and reporting.

In the Ecommerce Review Sentiment Analysis project, I used Python, and Streamlit to classify customer reviews as positive or negative using NLP techniques. The project also included real-time prediction and visualization using Streamlit.

I also gained experience in SQL and PostgreSQL using pgAdmin for database operations such as joins, filtering, aggregations, and queries. In addition, I learned Power BI dashboard creation and reporting techniques.

The internship also introduced me to Deep Learning concepts such as Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and image preprocessing using TensorFlow and Keras.

Overall, the internship improved my technical knowledge, analytical thinking, problem-solving skills, and confidence in handling real-world Data Science and Machine Learning projects.

5.2 Project Outputs

The Pizza Sales Analysis project analyzed sales performance and customer ordering patterns using Python, SQL, and Power BI. The project generated outputs such as total revenue, total orders, average order value, sales trends, and best-selling pizzas. Interactive dashboards and visual reports were created for business analysis.

The Ecommerce Review Sentiment Analysis project analyzed customer reviews using NLP and the DistilBERT model. The system classified reviews as positive or negative and displayed predictions with confidence scores through a Streamlit web application. Sentiment visualization charts were also generated.

Overall, both projects successfully produced meaningful analytical insights, visual reports, and prediction outputs using real-world datasets and Machine Learning techniques. These projects demonstrated practical applications of Data Science, NLP, SQL, Business Analytics, and dashboard visualization in solving real-world business problems.

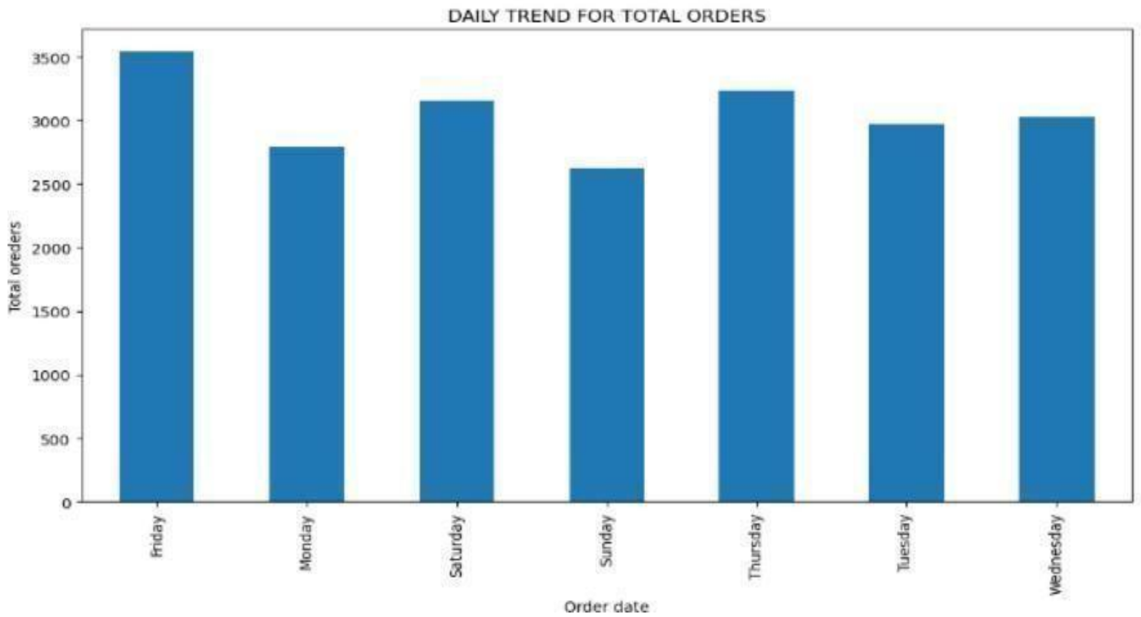
5.2.1 Pizza Sales Analysis System

The interface represents the visualization module of the Pizza Sales Analysis System. The bar chart illustrates the daily trend for total orders by analyzing customer purchase data grouped according to weekdays using Python libraries such as Pandas and Matplotlib.

```
daily_orders = df.groupby('day_of_week')['order_id'].nunique()

plt.figure(figsize=(12,6))
daily_orders.plot(kind='bar')

plt.title("DAILY TREND FOR TOTAL ORDERS")
plt.xlabel("Order date")
plt.ylabel("Total oreders")
plt.show()
```



Snapshot 5.1: Pizza Sales Daily Order Trend Visualization

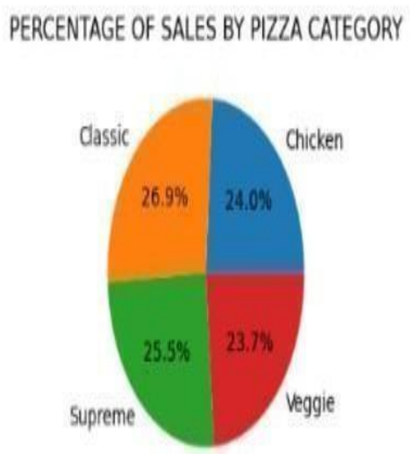
The visualization was created after preprocessing and organizing the sales dataset to better understand customer ordering behavior and sales distribution across the week.

The graph helps identify customer ordering patterns, peak business days, and overall sales trends. From the analysis, Friday recorded the highest number of total orders, indicating increased customer activity before weekends, while Sunday showed comparatively lower sales performance. The visualization also highlights how customer demand varies on different days, which is useful for understanding business performance and operational planning.

The analysis can support decision-making in areas such as inventory management, staff allocation, promotional activities, and sales strategy planning. By identifying peak order days and customer demand patterns, businesses can improve service efficiency, manage resources effectively, and enhance customer satisfaction. The visualization provides a clear and easy-to-understand representation of sales trends, making business analysis more efficient and data-driven.

```
#MONTHLY TREND FOR TOTAL ORDERS
import seaborn as sns
df['day_of_week']= df['order_date'].dt.month_name()
df['day_of_week'].unique()
```

```
array(['January', 'February', 'March', 'April', 'May', 'June', 'July',
       'August', 'September', 'October', 'November', 'December'],
      dtype=object)
```



Snapshot 5.2: Percentage of Sales by Pizza Category

The screenshot represents the category-wise sales distribution analysis in the Pizza Sales Analysis Dashboard. The chart displays the percentage contribution of each pizza category to the total sales revenue generated during the analysis period. The dataset was analyzed using Power BI and visualization techniques to understand how different pizza categories contribute to overall business performance.

Different pizza categories such as Classic, Supreme, Veggie, and Chicken were analyzed to identify customer preferences and the most popular categories based on sales percentage and order trends. The visualization clearly shows which category generated higher revenue and which categories had lower customer demand. This helps in understanding customer buying behavior and sales distribution across product categories.

The dashboard provides an easy-to-understand visual representation that helps compare category-wise sales contribution and identify high-performing pizza categories. It supports business decision-making related to inventory management, product planning, promotional activities, pricing strategies, and marketing campaigns. By analyzing category-wise sales trends, businesses can focus more on customer-preferred products, improve sales strategies, and optimize overall business performance.

1. **TOTAL REVENUE: The sum of the total price of all pizza orders.**

   Query: SELECT SUM(total_price) AS Total_Revenue FROM pizza_sales;

Results | Messages

	Total_Revenue
1	817860.05083847

2. **TOTAL ORDER"S: The total number of orders placed.**

   Query: SELECT COUNT(DISTINCT order_id) AS Total_Orders FROM pizza_sales;

Results | Messages

	Total_Orders
1	21350

3. **TOTAL PIZZA'S SOLD: The sum of the quantities of all pizas sold**

   Query: SELECT SUM(quantity) AS Total_Pizzas_Sold FROM pizza_sales;

Results | Messages

	Total_Pizzas_Sold
1	49574

Results | Messages

	pizza_name	revenue	total_quantity	total_orders
1	The Brie Carre Pizza	11588.4998130798	490	480
2	The Green Garden Pizza	13955.75	997	976
3	The Spinach Supreme Pizza	15277.75	950	918
4	The Mediterranean Pizza	15360.5	934	912
5	The Spinach Pesto Pizza	15596	970	945

8. Hourly Trend (Peak Hours)

Query: SELECT TOP 5
DATEPART(HOUR, order_time) AS hour,
COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY DATEPART(HOUR, order_time)
ORDER BY hour DESC;

Results | Messages

	hour	total_orders
1	23	28
2	22	663
3	21	1198
4	20	1642
5	19	2009

Snapshot 5.3: SQL Query Output for Pizza Sales Analysis

The screenshot represents the SQL query analysis performed in the Pizza Sales Analysis project. Different SQL aggregate functions such as SUM(), COUNT(), AVG(), and GROUP BY were used to calculate important business metrics including total revenue, total orders, total pizzas sold, and average order value. SQL queries were also written to analyze pizza-wise revenue, quantity sold, category-wise sales performance, and hourly sales trends during peak business hours.

The analysis involved retrieving and processing data from multiple tables using filtering, sorting, joins, grouping, and aggregation operations. Queries were used to identify the best-selling and least-selling pizzas based on revenue, quantity, and customer orders. Time-based analysis was also performed to understand peak ordering hours and daily sales trends.

The query results helped in identifying customer ordering behavior, sales performance patterns, and business growth trends. These insights supported better decision-making related to inventory management, staffing, marketing strategies, and sales planning. Overall, the SQL analysis demonstrated how database querying techniques can be effectively used to generate meaningful business insights from real-world sales data.

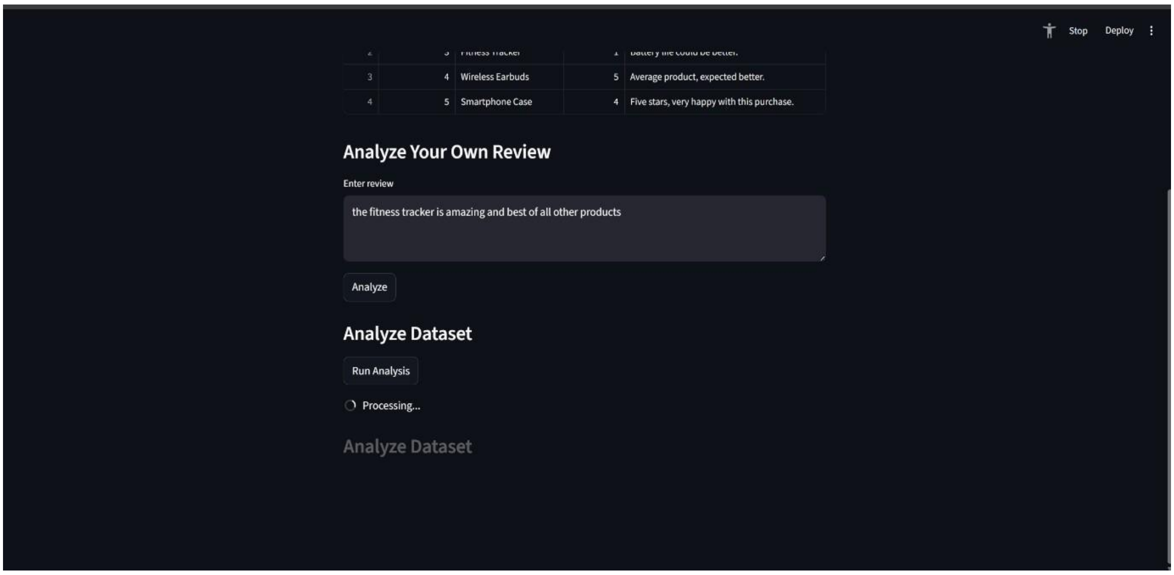
Snapshot 5.4: Pizza Sales Dash Board

The screenshot represents the SQL query analysis performed in the Pizza Sales Analysis project. Various SQL aggregate functions such as SUM(), COUNT(), AVG(), and GROUP BY were used to calculate important business metrics including total revenue, total orders, average order value, and total pizzas sold. Additional SQL queries were performed to analyze pizza-wise revenue, quantity sold, category-wise sales distribution, and hourly sales trends during peak business hours.

The queries helped in organizing and analyzing large sales datasets efficiently by filtering, grouping, and summarizing data based on business requirements. Different analytical queries were also used to identify top-selling pizzas, low-performing products, and customer ordering trends across different time periods.

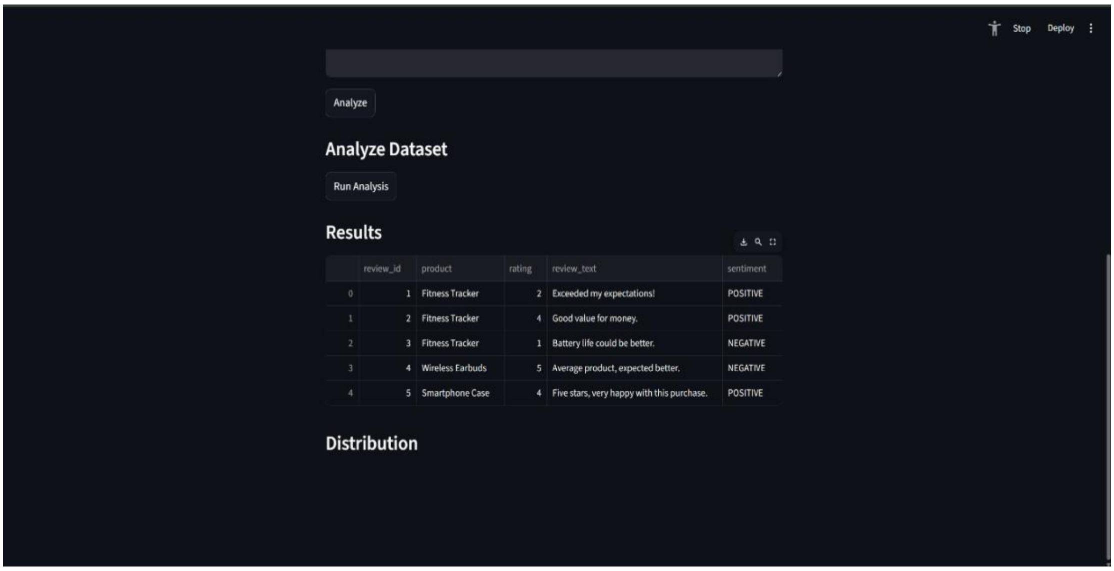
The query results provided meaningful insights into sales performance, customer purchasing behavior, and peak order timings, which supported effective business analysis and decision-making. This analysis can help improve inventory planning, sales strategies, customer service management, and overall business performance in the food and retail industry.

5.2.2 Ecommerce Review Sentiment Analysis



Snapshot 5.5: Ecommerce Review Input Analysis Interface

The screenshot shows the input interface of the Ecommerce Review Sentiment Analysis system developed using Machine Learning and Natural Language Processing (NLP) techniques. The system allows users to enter product reviews and analyze customer sentiment in real time. The application was developed using Python, Streamlit, and the DistilBERT Transformer model for sentiment classification. This interface demonstrates how user reviews are processed and analyzed to identify positive or negative sentiments effectively.



Snapshot 5.6: Ecommerce Review Sentiment Result Dashboard

The screenshot displays the sentiment analysis results generated from the ecommerce review dataset. The system processes customer reviews and classifies them into sentiments such as Positive and Negative using NLP and Machine Learning techniques. The dashboard presents review details, product information, ratings, review text, and predicted sentiment labels in a structured format for easier analysis and interpretation.

Snapshot 5.7: Ecommerce Review Sentiment Distribution Graph

The screenshot represents the sentiment distribution visualization generated from the ecommerce review dataset. The graph displays the number of Positive and Negative reviews identified by the sentiment analysis model. Visualization techniques were used to better understand customer feedback patterns and overall product satisfaction. This output helps in analyzing customer opinions and supports business decision-making through graphical representation of sentiment data.

CHAPTER 6

CONCLUSION

The internship at Take It Smart Pvt. Ltd. provided valuable practical exposure in the fields of Data Science, Machine Learning, Natural Language Processing, Data Analytics, and Business Intelligence. By working on projects such as Pizza Sales Analysis and Ecommerce Review Sentiment Analysis, I gained hands-on experience in handling real-world datasets, performing data preprocessing, exploratory data analysis, visualization, sentiment classification, and implementing machine learning techniques.

The Pizza Sales Analysis project demonstrated how Python, SQL, and Power BI can be used for business analytics, KPI analysis, sales trend identification, customer behavior analysis, and dashboard development to support data-driven decision-making in business environments. The Ecommerce Review Sentiment Analysis project highlighted the practical application of Natural Language Processing and Transformer-based models such as DistilBERT for analyzing customer reviews and generating sentiment predictions using real-time applications developed with Streamlit.

Apart from technical learning, the internship improved my analytical thinking, problem-solving abilities, communication skills, teamwork, and confidence in handling complete project workflows from data preprocessing to deployment. It also helped me understand professional work practices, reporting methods, dashboard creation, and the importance of data-driven solutions across different industries and domains.

Overall, the internship played an important role in strengthening my technical knowledge and preparing me for future opportunities in Data Science, Machine Learning, Artificial Intelligence, NLP, and Business Analytics with improved confidence and practical understanding.

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